EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Let $f(x) = [\tan^2 x]$, (where [.] denotes greatest integer function). Then -

(A) $\lim\limits_{x\to 0} f(x)$ does not exist (B) $f(x)$ is continuous at $x = 0$.

(C) $f(x)$ is not differentiable at $x = 0$ (D) $f'(0) = 1$

MDY001

2. Let f be differentiable at $x = 0$ and $f'(0) = 1$. Then $\lim\limits_{h\to 0} \dfrac{f(h) - f(-2h)}{h} =$

(A) 3 (B) 2 (C) 1 (D) –1

MDY002

3. Let $g(x) = \begin{cases} 3x^2 - 4\sqrt{x} + 1 & \text{for } x < 1 \\ ax + b & \text{for } x \geq 1 \end{cases}$.

If $g(x)$ is continuous and differentiable at $x = 1$ then -

(A) $a = b = 4$ (B) $a = b = -4$

(C) $a = 4$ and $b = -4$ (D) $a = -4$ and $b = 4$

MDY003

4. The function $g(x) = \begin{cases} x + b, & x < 0 \\ \cos x, & x \geq 0 \end{cases}$ can be made differentiable at $x = 0$ -

(A) if b is equal to zero (B) if b is not equal to zero

(C) if b takes any real value (D) for no value of b

Packers Academy JEE

MDY004

5. Which one of the following functions is continuous everywhere in its domain but has atleast one point where it is not differentiable?

(A) $f(x) = x^{1/3}$ (B) $f(x) = \dfrac{|x|}{x}$ (C) $f(x) = e^{-x}$ (D) $f(x) = \tan x$

MDY005

6. If $f(x) = (x^5 + 1)|x^2 - 4x - 5| + \sin|x| + \cos(|x - 1|)$, then $f(x)$ is not differentiable at -

(A) 2 points (B) 3 points (C) 4 points (D) zero points

MDY006

7. Let $f(x) = \begin{cases} x^3 + 2x^2 & x \in \mathbb{Q} \\ -x^3 + 2x^2 + ax & x \notin \mathbb{Q} \end{cases}$, then the integral value of $'a'$ so that $f(x)$ is differentiable at $x = 1$, is

(A) 2 (B) 6 (C) 7 (D) not possible

MDY007

8. Let $\mathbb{R}$ be the set of real numbers and $f : \mathbb{R} \to \mathbb{R}$, be a differentiable function such that $|f(x) - f(y)| \leq |x - y|^3 \; \forall \; x, y \in \mathbb{R}$. If $f(10) = 100$, then the value of $f(20)$ is equal to -

(A) 0 (B) 10 (C) 20 (D) 100

MDY008

For More Join: @IITJEE_Advanced

9. The function $f(x) = (x^2-1)|x^2-3x+2| + \cos(|x|)$ is NOT differentiable at :

(A) –1 (B) 0 (C) 1 (D) 2

MDY009

10. If $f(x)=\begin{cases} e^{x^2}+x^3+1 & ; \quad x>0 \\ ax^2+bx+2 & ; \quad x\le 0 \end{cases}$ is differentiable at $x = 0$, then -

(A) $a \in R$ & $b = 0$ (B) $a \in R$ & $b = 1$ (C) $a \in R$ & $b = 2$ (D) $a \in R$ & $b = 3$

MDY010

11. If $f(x) = [\sin x] + \sqrt{\sin x - [\sin x]}$, where [.] denotes greatest integer function, then $f(x)$ is -

(A) differentiable at $x=\frac{\pi}{2}$ & $x = \pi$ both

(B) differentiable at $x=\frac{\pi}{2}$ but not at $x = \pi$

(C) differentiable at $x = \pi$ but not at $x=\frac{\pi}{2}$

(D) neither differentiable at $x = \pi$ nor at $x=\frac{\pi}{2}$

MDY011

12. If $f(x)=\begin{cases} \sin^{-1}x+a\cos\pi x & -1<x<0 \\ b\cos^{-1}x+\sin\pi x & 0\le x<1 \end{cases}$ is differentiable in $(-1, 1)$, then $(2a + b)$ is equal to -

(A) π^2 (B) $1+\pi^2$ (C) $1-\pi^2$ (D) π^2-1

MDY012

Rankers Academy JEE

13. Number of points where $f(x) = |x - sgn(x)|$ is non-differentiable, is (sgn(.) denotes signum function)

(A) 0 (B) 1 (C) 2 (D) 3

MDY152

14. Let $f(x)=\begin{cases} \frac{\ell n\cos x}{ax} & , \quad x>0 \\ 0 & , \quad x=0 \\ \frac{e^{x^2}-1}{bx} & , \quad x<0 \end{cases}$. If $f'(0)=\frac{1}{4}$, then

(A) $a + b = 2$ (B) $b - a = 2$ (C) $a + b = -2$ (D) $b - a = 4$

MDY153

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If $f(x) = x\left(\sqrt{x} - \sqrt{x+1}\right)$, then -

(A) $Rf'(0)$ exist

(B) $L\ f'(0)$ exist but $R\ f'(0)$ does not exist

(C) $\lim\limits_{x\to 0^+} f(x)$ exist

(D) $f(x)$ is differentiable at $x = 0$

MDY041

2. The function $f(x)=\begin{cases}|x-3|, x\geq 1\\ \left(\frac{x^2}{4}\right)-\left(\frac{3x}{2}\right)+\left(\frac{13}{4}\right), x<1\end{cases}$ is -

(A) continuous at $x = 1$

(B) differentiable at $x = 1$

(C) continuous at $x = 3$

(D) differentiable at $x = 3$

MDY042

3. Select the correct statements -

(A) The function f defined by $f(x)=\begin{cases}2x^2+3 & \text{for} \quad x\leq 1\\ 3x+2 & \text{for} \quad x>1\end{cases}$ is neither differentiable nor continuous at $x = 1$.

(B) The function $f(x) = x^2\ |x|$ is twice differentiable at $x = 0$.

(C) If f is continuous at $x = 5$ and $f(5) = 2$ then $\lim\limits_{x\to 2} f(4x^2 - 11)$ exists

(D) If $\lim\limits_{x\to a}\left(f(x)+g(x)\right)=2$ and $\lim\limits_{x\to a}\left(f(x)-g(x)\right)=1$ then $\lim\limits_{x\to a} f(x).g(x)$ need not exist.

Rankers Academy JEE

MDY043

4. If $f(x) = \text{sgn}(x^5)$, then which of the following is/are **false** (where sgn denotes signum function) -

(A) $f'(0^+) = 1$

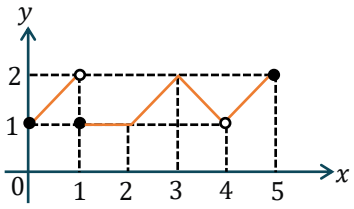
(B) $f'(0^-) = -1$

(C) f is continuous but not differentiable at $x = 0$

(D) f is discontinuous at $x = 0$

MDY044

5. Graph of $f(x)$ is shown in adjacent figure, then



(A) $f(x)$ has non-removable discontinuity at two points

(B) $f(x)$ is non-differentiable at three points in its domain

(C) $\lim\limits_{x\to 1} f(f(x))=1$

(D) Number of points of discontinuity = number of points of non-differentiability

MDY045

6. Let S denotes the set of all points where $\sqrt[5]{x^2\,|x|^3} - \sqrt[3]{x^2\,|x|} - 1$ is not differentiable then S is a subset of -

(A) {0, 1}

(B) {0, 1, –1}

(C) {0, 1}

(D) {0}

MDY046

For More Join: @IITJEE_Advanced

7. Which of the following statements is/are correct ?

(A) There exists a function $f : [0,1] \to \mathbb{R}$ which is discontinuous at every point in [0,1] & $|f(x)|$ is continuous at every point in [0,1]

(B) Let $F(x) = f(x).g(x)$. If $f(x)$ is differentiable at $x = a, f(a) = 0$ and $g(x)$ is continuous at $x = a$ then $F(x)$ is always differentiable at $x = a$.

(C) If $Rf'(a) = 2$ & $Lf'(a) = 3$, then $f(x)$ is non-differentiable at $x = a$ but will be always continuous at $x = a$

(D) If $f(a)$ and $f(b)$ possess opposite signs then there must exist at least one solution of the equation $f(x) = 0$ in (a, b) provided f is continuous on $[a, b]$

MDY047

8. Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) = |f(x)|$ for all x. Then which of the following is/are not always true-

(A) If f is continuous then g is also continuous

(B) If f is one-one then g is also one-one

(C) If f is onto then g is also onto

(D) If f is differentiable then g is also differentiable

MDY048

9. The function $\phi(x) = [|x| - \sin|x|]$ (where [.] denotes greater integer function) is -

(A) derivable at $x = 0$ (B) continuous at $x = 0$

(C) $\lim\limits_{x\to 0}\phi(x)$ does not exists (D) continuous and derivable at $x = 0$

MDY049

10. Let $f(x) = \begin{cases} x^2\cos\frac{1}{x}, & x<0 \\ 0, & x=0 \\ x^2\sin\frac{1}{x}, & x>0 \end{cases}$, then which of the following is (are) correct?

(A) $f(x)$ is continuous but not differentiable at $x = 0$

(B) $f(x)$ is continuous and differentiable at $x = 0$

(C) $f'(x)$ is continuous but not differentiable at $x = 0$

(D) $f'(x)$ is discontinuous at $x = 0$

MDY050

11. $f(x) = \begin{cases} a+\sin(\sin x); & x \geq 0 \\ \ell n(\cos x)+bx; & x<0 \end{cases}$ is differentiable at $x = 0$, then -

(A) $a = 1$ (B) $b = 0$ (C) $a = 0$ (D) $b = 1$

MDY051

12. Let $f(x + y) + f(x - y) = 2f(x)$ $\forall x, y \in R$, then which of the following is/are be correct ($f(0) = 0$ & f is differentiable $\forall\, x \in \mathbb{R}$)

(A) f must be odd (B) f must be even (C) $f'(1) = f'(2)$ (D) $f'(2) = f'(3) = f'(4)$

MDY052

13. Let $f(x) = \begin{cases} 1-x, & (0 \leq x \leq 1) \\ x+2, & (1<x<2) \\ 4-x, & (2 \leq x \leq 4) \end{cases}$. Then in $0 \leq x \leq 4$ function $y = f(f(x))$ is

(A) continuous at $x = 1$

(B) not differentiable at $x = 1$

(C) continuous and differentiable at $x = 2$

(D) continuous but not differentiable at $x = 3$

MDY053

Rankers Academy JEE

For More Join: @IITJEE_Advanced

14. If $\lim\limits_{x\to 0}\dfrac{1-\cos\left(1-\cos\dfrac{x}{2}\right)}{2^m x^n}$ is equal to the left hand derivative of $e^{-|x|}$ at $x=0$, then the value of $|n+2m|$ is divisible by

(A) 2 (B) 3 (C) 5 (D) 7

MDY054

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 15 to 16

A new function tignum (x) (i.e. tgn(x)) is defined such that tgn $(x)=\begin{cases}1 & \text{if } [x]=\text{even}\\ -1 & \text{if } [x]=\text{odd}\end{cases}$,

where [.] denotes greatest integer function. Let $f(x)=\text{tgn}(x)\sin(x)|x|$.

On the basis of above information, answer the following questions :

15. Number of points in [0, 10] where $f(x)$ is discontinuous is -

(A) 8 (B) 9 (C) 10 (D) 11

MDY055

16. Number of points in $[-10,10]$ where $f(x)$ is non- derivable is -

(A) 18 (B) 19 (C) 20 (D) 21

MDY056

Paragraph for Question No. 17 to 18

Consider the function $f(x)=\begin{cases}x^2-5x+6 & ; \ -2\le x\le 4\\ k-\tan\left(\dfrac{\pi x}{4}\right) & ; \ 4<x\le 5\\ \log_{10}(\alpha x) & ; \ 5<x\le 10\end{cases}$. Let $f_1(x)=|f(|x|)|$.

On the basis of above information, answer the following questions :

17. If $f(x)$ is continuous in $[-2,10]$ then -

(A) $\alpha=2$ (B) $\alpha=4$ (C) $k=2$ (D) $k=4$

MDY057

18. If $f(x)$ is continuous in $[-2,10]$, then the number of points, where $f_1(x)$ is NOT differentiable, is -

(A) 9 (B) 8 (C) 7 (D) 5

MDY058

MATRIX MATCH TYPE QUESTION

19.

	List-I		List-II
(P)	If $f(x)$ is derivable at $x=3$ & $f'(3)=2$, then $\lim\limits_{h\to 0}\dfrac{f(3+h^2)-f(3-h^2)}{2h^2}$ equals	(1)	0
(Q)	Let $f(x)$ be a function satisfying the condition $f(-x)=f(x)$ for all real x. If $f'(0)$ exists, then its value is equal to	(2)	1
(R)	For the function $f(x)=\begin{cases}\dfrac{x}{1+e^{1/x}}, & x\neq 0\\ 0, & x=0\end{cases}$ the derivative from the left $Lf'(0)$ equals	(3)	2
(S)	The number of points at which the function $f(x)=\max.\{a-x, a+x, b\}, -\infty<x<\infty$, $0<a<b$ cannot be differentiable is	(4)	3

(A) (P) $\to$ 3, (Q) $\to$ 2, (R) $\to$ 1, (S) $\to$ 3 (B) (P) $\to$ 4, (Q) $\to$ 1, (R) $\to$ 2, (S) $\to$ 4

(C) (P) $\to$ 3, (Q) $\to$ 1, (R) $\to$ 2, (S) $\to$ 3 (D) (P) $\to$ 2, (Q) $\to$ 2, (R) $\to$ 1, (S) $\to$ 3

MDY059

For More Join: @IITJEE_Advanced

EXERCISE (S)

1. If the function $f(x)$ defined as $f(x)=\begin{cases} -\frac{x^2}{2} & \text{for} \quad x\le 0 \\ x^n \sin\frac{1}{x} & \text{for} \quad x>0 \end{cases}$ is continuous but not derivable at $x=0$ then sum of all non-negative integral value(s) of 'n' is

MDY013

2. If $f(x)=|x-1|.([x]-[-x])$, where $[x]$ denotes greatest integer function, then the value of $(Rf'(1)+Lf'(1))$ is

MDY014

3. For any real number x, let $[x]$ denote the largest integer less than or equal to x. Let f be a real valued function defined on the interval $[-3,3]$ by $f(x)=\begin{cases} -x-[-x] & \text{if} \quad [x] \text{ is even} \\ x-[x] & \text{if} \quad [x] \text{ is odd} \end{cases}$

If L denotes the number of point of discontinuity and M denotes the number of points of non-derivability of $f(x)$, then find $(L+M)$.

MDY015

4. A derivable function $f:\mathbb{R}^+\to\mathbb{R}$ satisfies the condition $f(x)-f(y)\ge \ell n\,(x/y)+x-y$ for every $x,y\in\mathbb{R}^+$. If g denotes the derivative of f then compute the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$.

Rankers Academy JEE

MDY016

5. Let $f(x)=[[x]+\{x^2\}]+\{[x^2]+\{x\}\}$, then number of points where $|f(x)|$ is non-derivable in $[-3,3]$ is equal to (where [.] denotes greatest integer function & {.} denotes fractional part function)

MDY017

6. If f is a differentiable function such that $f\left(\frac{x+y}{3}\right)=\frac{f(x)+f(y)+f(0)}{3}$, $\forall\, x,y\in\mathbb{R}$ and $f'(0)=2$, then the slope of tangent to $f(x)$ at $x=7$ is

MDY018

7. Let $f(x)$ be a differentiable function such that $\lim_{x\to 0}\frac{f(3-\sin x)-f(3+x)}{x}=8$, then $|f'(3)|$ is

MDY019

8. Let $f(x)$ be a differentiable function such that $2f(x+y)+f(x-y)=3f(x)+3f(y)+2xy$ $\forall\, x,y\in\mathbb{R}$ & $f'(0)=0$, then $f(10)+f'(10)$ is equal to

MDY020

9. Let $[x]$ denote the greatest integer less than or equal to x. The number of integral points in $[-1,1]$ where $f(x)=[x\sin\pi x]$ is differentiable are

MDY130

10. Find the natural number 'a' for which $\sum_{k=1}^{n} f(a+k)=2048(2^n-1)$, where the function '$f$' satisfies the relation $f(x+y)=f(x).f(y)$ for all natural numbers x & y and further $f(1)=2$

MDY131

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Let f be a differentiable function from $\mathbb{R}$ to $\mathbb{R}$ such that $|f(x)-f(y)| \le 2|x-y|^{\frac{3}{2}}$, for all $x, y \in \mathbb{R}$. If $f(0)=1$ then $\int_0^1 f^2(x)dx$ is equal to **[JEE (Main) 2019]**

(1) 0 (2) $\frac{1}{2}$ (3) 2 (4) 1

MDY022

2. Let $f(x)=\begin{cases}\max\{|x|,x^2\}, |x| \le 2 \\ 8-2|x|, 2<|x|\le 4\end{cases}$

Let S be the set of points in the interval $(-4,4)$ at which f is not differentiable. Then S: **[JEE (Main) 2019]**

(1) is an empty set (2) equals $\{-2,-1,1,2\}$
(3) equals $\{-2,-1,0,1,2\}$ (4) equals $\{-2,2\}$

MDY023

3. Let $f(x)=\begin{cases}-1, -2 \le x < 0 \\ x^2-1, 0 \le x \le 2\end{cases}$ and $g(x)=|f(x)|+f(|x|)$. Then, in the interval $(-2,2)$, g is:- **[JEE (Main) 2019]**

(1) differentiable at all points (2) not differentiable at two points
(3) not continuous (4) not differentiable at one point

MDY027

4. Let S be the set of points where the function, $f(x)=|2-|x-3||, x \in \mathbb{R}$, is not differentiable. Then $\sum_{x \in S} f(f(x))$ is equal to _____. **[JEE (Main) 2020]**

Rankers Academy JEE

MDY030

5. Suppose a differentiable function $f(x)$ satisfies the identity $f(x+y)=f(x)+f(y)+xy^2+x^2y$, for all real x and y. If $\lim_{x \to 0} \frac{f(x)}{x}=1$, then $f'(3)$ is equal to. **[JEE (Main) 2020]**

MDY031

6. The function $f(x)=\begin{cases}\frac{\pi}{4}+\tan^{-1}x, |x| \ge 1 \\ \frac{1}{2}(|x|-1), |x|<1\end{cases}$ is : **[JEE (Main) 2020]**

(1) continuous on $\mathbb{R}-\{1\}$ and differentiable on $\mathbb{R}-\{-1,1\}$.
(2) both continuous and differentiable on $\mathbb{R}-\{-1\}$.
(3) continuous on $\mathbb{R}-\{-1\}$ and differentiable on $\mathbb{R}-\{-1,1\}$.
(4) both continuous and differentiable on $\mathbb{R}-\{1\}$

MDY032

7. If the function $f(x)=\begin{cases}k_1(x-\pi)^2-1, x \le \pi \\ k_2\cos x, \quad x>\pi\end{cases}$ is twice differentiable, then the ordered pair (k_1,k_2) is equal to : **[JEE (Main) 2020]**

(1) $\left(\frac{1}{2},1\right)$ (2) $(1,1)$ (3) $\left(\frac{1}{2},-1\right)$ (4) $(1,0)$

MDY033

For More Join: @IITJEE_Advanced

8. Let $f:\mathbb{R}\to\mathbb{R}$ be defined as $f(x)=\begin{cases}x^5\sin\left(\frac{1}{x}\right)+5x^2 & , \quad x<0\\ 0 & , \quad x=0.\\ x^5\cos\left(\frac{1}{x}\right)+\lambda x^2 & , \quad x>0\end{cases}$

The value of λ for which $f''(0)$ exists, is. **[JEE (Main) 2020]**

MDY034

9. Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined by $f(x)=\max\{x,x^2\}$. Let S denote the set of all points in $\mathbb{R}$, where f is not differentiable. Then: **[JEE (Main) 2020]**

(1) $\{0,1\}$ (2) $\{0\}$ (3) ϕ (an empty set) (4) $\{1\}$

MDY035

10. Let a function $g:[0,4]\to\mathbb{R}$ be defined as $g(x)=\begin{cases}\max\limits_{0\le t\le x}\{t^3-6t^2+9t-3\} & , \ 0\le x\le 3\\ 4-x & , \ 3<x\le 4\end{cases}$,

then the number of points in the interval $(0,4)$ where $g(x)$ is NOT differentiable, is _____.

[JEE (Main) 2021]

MDY036

11. Let $f:\mathbb{R}\to\mathbb{R}$ be a function defined as

$$f(x)=\begin{cases}3\left(1-\frac{|x|}{2}\right) & \text{if} \quad |x|\le 2\\ 0 & \text{if} \quad |x|>2\end{cases}$$

Let $g:\mathbb{R}\to\mathbb{R}$ be given by $g(x)=f(x+2)-f(x-2)$. If n and m denote the number of points in $\mathbb{R}$ where g is not continuous and not differentiable, respectively, then $n+m$ is equal to _____.

[JEE (Main) 2021]

MDY037

12. The number of points, at which the function $f(x)=|2x+1|-3|x+2|+|x^2+x-2|, x\in\mathbb{R}$ is not differentiable, is _____. **[JEE (Main) 2021]**

MDY038

13. Let $[t]$ denote the greatest integer less than or equal to t. Let $f(x)=x-[x]$, $g(x)=1-x+[x]$, and $h(x)=\min\{f(x),g(x)\}, x\in[-2,2]$. Then h is : **[JEE (Main) 2021]**

(1) continuous in [–2, 2] but not differentiable at more than four points in (–2, 2)

(2) not continuous at exactly three points in [–2, 2]

(3) continuous in [–2, 2] but not differentiable at exactly three points in (–2, 2)

(4) not continuous at exactly four points in [–2, 2]

MDY039

14. The function $f(x)=|x^2-2x-3|\cdot e^{|9x^2-12x+4|}$ is not differentiable at exactly : **[JEE (Main) 2021]**

(1) four points (2) three points (3) two points (4) one point

MDY040

15. If $[t]$ denotes the greatest integer $\le$ t, then number of points, at which the function

$f(x)=4|2x+3|+9\left[x+\frac{1}{2}\right]-12[x+20]$ is not differentiable in the open interval (–20, 20), is.

[JEE (Main) 2022]

MDY073

For More Join: @IITJEE_Advanced

16. Let $f(x) = \min\{1, 1 + x\sin x\}$, $0 \le x \le 2\pi$. If m is the number of points, where f is not differentiable and n is the number of points, where f is not continuous, then the ordered pair (m, n) is equal to

[JEE (Main) 2022]

(1) (2, 0) (2) (1, 0) (3) (1, 1) (4) (2, 1)

MDY074

17. Let f: (-2, 2) $\to$ IR be defined by

$$f(x)=\begin{cases} x[x] & ,-2<x<0 \\ (x-1)[x] & ,0\le x<2 \end{cases}$$

Where $[x]$ denotes the greatest integer function. If m and n respectively are the number of points in (-2, 2) at which $y = |f(x)|$ is not continuous and not differentiable, then $m + n$ is equal to _____.

[JEE (Main) 2023]

MDY075

18. Let $[x]$ denote the greatest integer function and $f(x) = \max\{1 + x + [x], 2 + x, x + 2[x]\}$, $0 \le x \le 2$. Let m be the number of points in [0,2], where f is not continuous and n be the number of points in (0,2), where f is not differentiable. Then $(m + n)^2 + 2$ is equal to: **[JEE (Main) 2023]**

(1) 11 (2) 2 (3) 6 (4) 3

MDY076

19. Let $f(x)=\begin{cases} x^2\sin\left(\frac{1}{x}\right) & ,x\neq 0 \\ 0 & ,x=0 \end{cases}$; Then at $x = 0$ **[JEE (Main) 2023]**

(1) f is continuous but not differentiable

(2) f is continuous but f' is not continuous

(3) f and f' both are continuous

(4) f' is continuous but not differentiable

MDY077

20. Let $f:\mathbb{R}-\{0\}\to\mathbb{R}$ be a function satisfying $f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}$ for all $x, y, f(y) \neq 0$. If $f'(1) = 2024$, then

[JEE (Main) 2024]

(1) $xf'(x) - 2024 f(x) = 0$ (2) $xf'(x) + 2024f(x) = 0$

(3) $xf'(x) + f(x) = 2024$ (4) $xf'(x) - 2023f(x) = 0$

MDY128

21. If the function $f(x)=\begin{cases} \frac{1}{|x|} & ,|x|\ge 2 \\ ax^2+2b, & |x|<2 \end{cases}$ is differentiable on R, then $48(a + b)$ is equal to ____.

[JEE (Main) 2024]

MDY129

Bansal's Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Let $f(x)=\begin{cases} x^2\left|\cos\frac{\pi}{x}\right| & , \quad x\neq 0 \\ 0 & , \quad x=0 \end{cases}$, $x \in R$, then f is - **[JEE (Advanced) 2012]**

(A) differentiable both at $x = 0$ and at $x = 2$
(B) differentiable at $x = 0$ but not differentiable at $x = 2$
(C) not differentiable at $x = 0$ but differentiable at $x = 2$
(D) differentiable neither at $x = 0$ nor at $x = 2$

MDY061

2. Let $f_1 : \mathbb{R} \to \mathbb{R}$, $f_2 : [0, \infty) \to \mathbb{R}$, $f_3 : \mathbb{R} \to \mathbb{R}$ and $f_4 : \mathbb{R} \to [0, \infty)$ be defined by

$$f_1(x) = \begin{cases} |x| & \text{if} \quad x<0, \\ e^x & \text{if} \quad x\geq 0; \end{cases}$$

$$f_2(x) = x^2 \; ; \; f_3(x) = \begin{cases} \sin x & \text{if} \quad x<0, \\ x & \text{if} \quad x\geq 0 \end{cases}$$

and $f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if} \quad x<0, \\ f_2(f_1(x))-1 & \text{if} \quad x\geq 0. \end{cases}$

	List-I		**List-II**
P.	f_4 is	1.	onto but not one-one
Q.	f_3 is	2.	neither continuous nor one-one
R.	$f_2 o f_1$ is	3.	differentiable but not one-one
S.	f_2 is	4.	continuous and one-one

Rankers Academy JEE

[JEE (Advanced) 2014]

Codes :

	P	Q	R	S
(A)	3	1	4	2
(B)	1	3	4	2
(C)	3	1	2	4
(D)	1	3	2	4

MDY062

3. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define

$$h : \mathbb{R} \to \mathbb{R} \text{ by } h(x)=\begin{cases} \max\{f(x), g(x)\} & \text{if} \quad x\leq 0, \\ \min\{f(x), g(x)\} & \text{if} \quad x>0. \end{cases}$$

The number of points at which $h(x)$ is not differentiable is **[JEE (Advanced) 2014]**

MDY063

4. Let $a, b \in \mathbb{R}$ and $f: \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = a\cos(|x^3 - x|) + b|x|\sin(|x^3 + x|)$. Then f is -

(A) differentiable at $x = 0$ if $a = 0$ and $b = 1$
(B) differentiable at $x = 1$ if $a = 1$ and $b = 0$
(C) **NOT** differentiable at $x = 0$ if $a = 1$ and $b = 0$
(D) **NOT** differentiable at $x = 1$ if $a = 1$ and $b = 1$ **[JEE (Advanced) 2016]**

MDY064

For More Join: @IITJEE_Advanced

5. Let $f:\left[-\frac{1}{2},2\right]\to\mathbb{R}$ and $g:\left[-\frac{1}{2},2\right]\to\mathbb{R}$ be function defined by $f(x)=[x^2-3]$ and $g(x)=|x|f(x)+|4x-7|\,f(x)$, where $[y]$ denotes the greatest integer less than or equal to y for $y\in\mathbb{R}$. Then

(A) f is discontinuous exactly at three points in $\left[-\frac{1}{2},2\right]$

(B) f is discontinuous exactly at four points in $\left[-\frac{1}{2},2\right]$

(C) g is NOT differentiable exactly at four points in $\left(-\frac{1}{2},2\right)$

(D) g is NOT differentiable exactly at five points in $\left(-\frac{1}{2},2\right)$ **[JEE (Advanced) 2016]**

MDY065

6. Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function with $f(0)=1$ and satisfying the equation $f(x+y)=f(x)f'(y)+f'(x)f(y)$ for all $x,y\in\mathbb{R}$. Then, then value of $\log_e(f(4))$ is ____ .

[JEE (Advanced) 2018]

MDY066

7. Let $f_1:\mathbb{R}\to\mathbb{R}$, $f_2:\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\to\mathbb{R}$, $f_3:\left(-1,e^{\frac{\pi}{2}}-2\right)\to\mathbb{R}$ and $f_4:\mathbb{R}\to\mathbb{R}$ be functions defined by

(i) $f_1(x)=\sin\left(\sqrt{1-e^{-x^2}}\right)$

(ii) $f_2(x)=\begin{cases}\frac{|\sin x|}{\tan^{-1}x} & \text{if } x\neq 0\\ 1 & \text{if } x=0\end{cases}$, where the inverse trigonometric function $\tan^{-1}x$ assumes values in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$,

(iii) $f_3(x)=[\sin(\log_e(x+2)]$, where for $t\in\mathbb{R}$, $[t]$ denotes the greatest integer less than or equal to t,

(iv) $f_4(x)=\begin{cases}x^2\sin\left(\frac{1}{x}\right) & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$

	List-I		List-II
P.	The function f_1 is	**1.**	**NOT** continuous at $x=0$
Q.	The function f_2 is	**2.**	continuous at $x=0$ and **NOT** differentiable at $x=0$
R.	The function f_3 is	**3.**	differentiable at $x=0$ and its derivative is **NOT** continuous at $x=0$
S.	The function f_4 is	**4.**	differentiable at $x=0$ and its derivative is continuous at $x=0$

[JEE (Advanced) 2018]

The correct option is :

(A) P $\to$ 2; Q $\to$ 3, R $\to$ 1; S $\to$ 4

(B) P $\to$ 4; Q $\to$ 1; R $\to$ 2; S $\to$ 3

(C) P $\to$ 4; Q $\to$ 2, R $\to$ 1; S $\to$ 3

(D) P $\to$ 2; Q $\to$ 1; R $\to$ 4; S $\to$ 3

MDY067

Rankers Academy JEE
For More Join: @IITJEE_Advanced

8. Let the function $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3 - x^2 + (x-1)\sin x$ and let $g:\mathbb{R} \to \mathbb{R}$ be an arbitrary function. Let $fg: \mathbb{R} \to \mathbb{R}$ be the product function defined by $(fg)(x) = f(x)g(x)$. Then which of the following statements is/are TRUE ? **[JEE (Advanced) 2020]**

(A) If g is continuous at $x = 1$, then fg is differentiable at $x = 1$

(B) If fg is differentiable at $x = 1$, then g is continuous at $x = 1$

(C) If g is differentiable at $x = 1$, then fg is differentiable at $x = 1$

(D) If fg is differentiable at $x = 1$, then g is differentiable at $x = 1$

MDY068

9. Let $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ be functions satisfying $f(x+y) = f(x) + f(y) + f(x)f(y)$ and $f(x) = xg(x)$ for all $x, y \in \mathbb{R}$. If $\lim\limits_{x\to 0} g(x) = 1$, then which of the following statements is/are TRUE? **[JEE (Advanced) 2020]**

(A) f is differentiable at every $x \in \mathbb{R}$

(B) If $g(0) = 1$, then g is differentiable at every $x \in \mathbb{R}$

(C) The derivative $f'(1)$ is equal to 1

(D) The derivative $f'(0)$ is equal to 1

MDY069

10. Let the functions $f: (-1, 1) \to \mathbb{R}$ and $g: (-1,1) \to (-1, 1)$ be defined by

$f(x) = |2x - 1| + |2x + 1|$ and $g(x) = x - [x]$,

where $[x]$ denotes the greatest integer less than or equal to x. Let $fog: (-1, 1) \to \mathbb{R}$ be the composite function defined by $(fog)(x) = f(g(x))$. Suppose c is the number of points in the interval $(-1, 1)$ at which fog is **NOT** continuous, and suppose d is the number of points in the interval $(-1, 1)$ at which fog is **NOT** differentiable. Then the value of $c + d$ is ____

Rankers Academy JEE

[JEE (Advanced) 2020]

MDY070

11. Let $f : (0,1) \to \mathbb{R}$ be the function defined as $f(x) = [4x]\left(x - \frac{1}{4}\right)^2\left(x - \frac{1}{2}\right)$, where $[x]$ denotes the greatest integer less than or equal to x . Then which of the following statements is(are) true?

[JEE (Advanced) 2023]

(A) The function f is discontinuous exactly at one point in (0,1)

(B) There is exactly one point in (0,1) at which the function f is continuous but **NOT** differentiable

(C) The function f is **NOT** differentiable at more than three points in (0,1)

(D) The minimum value of the function f is $-\frac{1}{512}$

MDY071

12. Let $S = (0, 1) \cup (1, 2) \cup (3, 4)$ and $T = \{0, 1, 2, 3\}$. Then which of the following statements is(are) true ?

[JEE (Advanced) 2023]

(A) There are infinitely many functions from S to T

(B) There are infinitely many strictly increasing functions from S to T

(C) The number of continuous functions from S to T is at most 120

(D) Every continuous function from S to T is differentiable

MDY072

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	A	C	D	A	A	D	D	D	A
Que.	11	12	13	14						
Ans.	B	D	D	A						

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C	B,C	A,B,C	B,C	A,B,C,D	A,B,C,D	B,C,D	A,B,D	B,D
Que.	11	12	13	14	15	16	17	18	19	
Ans.	C,D	A,C,D	A,B	A,C	C	B	A,C	A	C	

EXERCISE (S)

1.	1	**2.**	2	**3.**	8	**4.**	5150	**5.**	1
6.	2	**7.**	4	**8.**	120	**9.**	3	**10.**	10

Rankers Academy JEE

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	4	3	10	1	1	5	1	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	1	3	79	2	4	4	2	1
Que.	21									
Ans.	15									

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	3	A,B	B,C	2	D	A,C	A,B,D	4
Que.	11	12								
Ans.	A,B	A,C,D								

For More Join: @IITJEE_Advanced